

INMO(10th) – 1995

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. In an acute angled triangle ABC, $m\angle A = 30^\circ$, H is the orthocenter and M is the midpoint of BC. On the line HM, take a point T such that $HM = MT$. Show that $AT = 2BC$.
 2. Show that there are infinitely many pairs (a, b) of relatively prime integers (not necessarily positive) such that both quadratic function $x^2 + ax + b = 0$ and $x^2 + 2ax + b = 0$ have integers roots.
 3. Show that the number of 3-elements {a, b, c} of $\{1, 2, 3, \dots, 63\}$ with $a + b + c < 95$ is less than the number of those with $a + b + c > 95$.
 4. Let ABC be a triangle and a circle Γ' be drawn inside the triangle, touching its incircle Γ externally and also touching the two sides AB and AC. Show that the ratio of the radii of the circles Γ' and Γ is equal to $\tan^2\left(\frac{\pi - A}{4}\right)$.
 5. Let $a_1, a_2, a_3, a_4, \dots, a_n$ be n real numbers all greater than 1 and such that $|a_k - a_{k+1}| < 1$ for $1 \leq k \leq n-1$. Show that $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \frac{a_3}{a_4} + \dots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 1$.
 6. Find all primes p for which the quotient $2^{p-1} - 1$ is a square.

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